

An Explainable Data-Driven Platform for Optimal Wind Farm Siting

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Submitted for FIT5126 as an exchange student at Monash University, May 2025

1 Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [1–3]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [4], reinforce this shift. These targets include:

- 40% of Victoria’s electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning ap-

proach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [5]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [6].

1.1 Aim

2 General Instructions

Manuscripts must be in two-column format. Exceptions to the two-column format include the title, authors’ names and complete

References

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